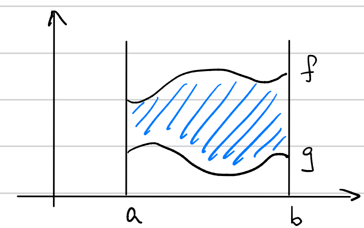
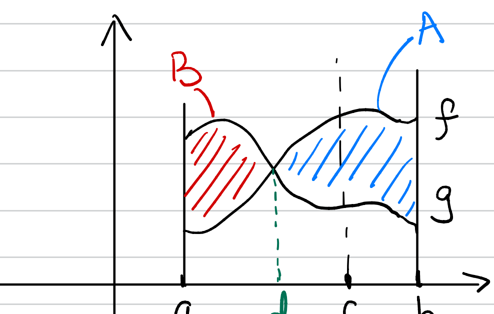
Area Between Curves

If f, g are continuous w/ $f(x) \geq g(x)$ throughout $[a, b]$, then the area of the region between the curves $y = f(x)$ and $y = g(x)$ from a to b is the integral

$$A = \int_a^b [f(x) - g(x)]\,dx$$





$$\int_a^b (f(x) - g(x))\,dx = A - B \neq \text{Area between curves}$$

$$\text{Area between curves} = A + B$$

$$\text{Area between curves}$$

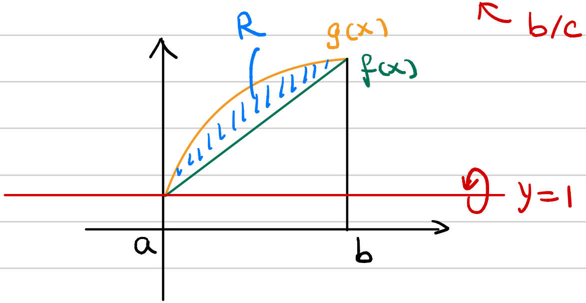
$$= \underbrace{\int_a^d (g(x) - f(x))\,dx}_{B} + \underbrace{\int_d^b (f(x) - g(x))\,dx}_{A}$$

Volumes by the Disk Method

f(x) revolved about x-axis

$$V = \pi \int_a^b (f(x) - \underset{\smile}{0})^2 \, dx$$

↖ b/c x-axis is y=0



region R revolved about y=1

g(x) first b/c g(x) is farther from y=1

$$V = \pi \int_a^b (\underline{g(x)} - 1)^2 - (f(x) - 1)^2 \, dx$$

b/c revolved about y=1

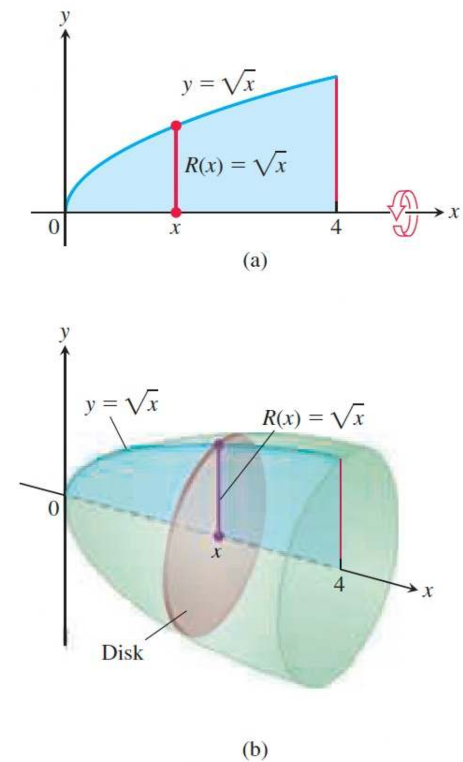


FIGURE 6.8 The region (a) and solid of revolution (b) in Example 4.

Volumes by the Shell Method

$$V = \int_a^b 2\pi \begin{pmatrix}\text{shell}\\ \text{radius}\end{pmatrix}\begin{pmatrix}\text{shell}\\ \text{height}\end{pmatrix} dx$$

Steps to think about:

① What is the interval of integration?
   $a = ?$ $b = ?$

② What is the shell radius?

③ What is the shell height?

If you don't know where to start, it is always helpful to first try to draw the cylindrical shell.

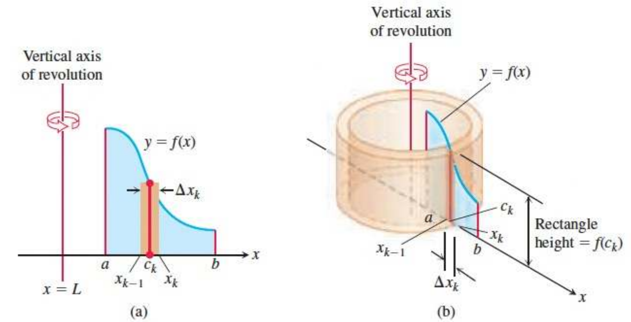


FIGURE 6.19 When the region shown in (a) is revolved about the vertical line $x = L$, a solid is produced which can be sliced into cylindrical shells. A typical shell is shown in (b).

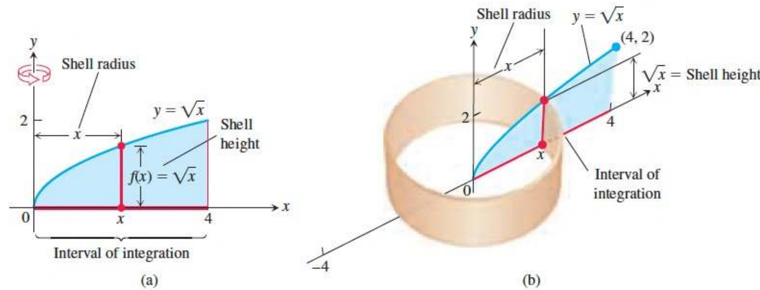


FIGURE 6.20 (a) The region, shell dimensions, and interval of integration in Example 2. (b) The shell swept out by the vertical segment in part (a) with a width Δx.

Integration by Parts

When we have $\int f(x)g(x)\,dx$ and u-sub doesn't work.

Goal: Convert $f(x)g(x)$ into one expression $h(x)$

Formula: $\int u\,dv = uv - \int v\,du$ $(*)$

Recipe: ① Figure out what are u and dv

e.g. $u = f(x)$ $dv = g(x)\,dx$

Normally: $f(x)$ is easier to differentiate
$g(x)$ is easier to integrate

② Compute du and v
e.g. $du = f'(x)\,dx$ $v = \int g(x)\,dx$

③ Plug in all these info (u, v, du, dv) into $(*)$

Powers and Products of Trig Functions

TRIG IDENTITIES

Reciprocal Identities

$\sin\theta = \frac{1}{\csc\theta}$ $\cos\theta = \frac{1}{\sec\theta}$ $\tan\theta = \frac{1}{\cot\theta}$

$\csc\theta = \frac{1}{\sin\theta}$ $\sec\theta = \frac{1}{\cos\theta}$ $\cot\theta = \frac{1}{\tan\theta}$

Quotient Identities

$\tan\theta = \frac{\sin\theta}{\cos\theta}$ $\cot\theta = \frac{\cos\theta}{\sin\theta}$

Pythagorean Identities

$1 = \sin^2\theta + \cos^2\theta$ $\sec^2\theta = \tan^2\theta + 1$ $\csc^2\theta = 1 + \cot^2\theta$

Power Reducing Identities

$\sin^2\theta = \frac{1-\cos 2\theta}{2}$ $\cos^2\theta = \frac{1+\cos 2\theta}{2}$ $\tan^2\theta = \frac{1-\cos 2\theta}{1+\cos 2\theta}$

DERIVATIVES

Trig Functions

$\frac{d}{dx}(\sin x) = \cos x$ $\frac{d}{dx}(\cos x) = -\sin x$ $\frac{d}{dx}(\tan x) = \sec^2 x$

$\frac{d}{dx}(\sec x) = \sec x \tan x$ $\frac{d}{dx}(\csc x) = -\csc x \cot x$ $\frac{d}{dx}(\cot x) = -\csc^2 x$

INTEGRALS

Trig Functions

$\int \sin u\,du = -\cos u + c$ $\int \cos u\,du = \sin u + c$ $\int \tan u\,du = \ln|\sec u| + c$

$\int \sec u\,du = \ln|\sec u + \tan u| + c$ $\int \csc u\,du = \ln|\csc u - \cot u| + c$ $\int \cot u\,du = \ln|\sin u| + c$

$\int \sec^2 u\,du = \tan u + c$ $\int \csc^2 u\,du = -\cot u + c$

Trigonometric Substitutions

TRIG IDENTITIES

Reciprocal Identities

$\sin\theta = \frac{1}{\csc\theta}$ $\cos\theta = \frac{1}{\sec\theta}$ $\tan\theta = \frac{1}{\cot\theta}$

$\csc\theta = \frac{1}{\sin\theta}$ $\sec\theta = \frac{1}{\cos\theta}$ $\cot\theta = \frac{1}{\tan\theta}$

Quotient Identities

$\tan\theta = \frac{\sin\theta}{\cos\theta}$ $\cot\theta = \frac{\cos\theta}{\sin\theta}$

Pythagorean Identities

$1 = \sin^2\theta + \cos^2\theta$ $\sec^2\theta = \tan^2\theta + 1$ $\csc^2\theta = 1 + \cot^2\theta$

$\sin^2(\theta) = 1 - \cos^2(\theta)$

$\cos^2(\theta) = 1 - \sin^2(\theta)$

trig sub: $x = \cos(\theta) \Rightarrow \sin^2(\theta) = 1 - x^2$

$x = \sin(\theta) \Rightarrow \cos^2(\theta) = 1 - x^2$

trig sub:

$x = \tan(\theta)$

$\Rightarrow \sec^2(\theta) = x^2 + 1$

DERIVATIVES

Trig Functions

$\frac{d}{dx}(\sin x) = \cos x$ $\frac{d}{dx}(\cos x) = -\sin x$ $\frac{d}{dx}(\tan x) = \sec^2 x$

$\frac{d}{dx}(\sec x) = \sec x \tan x$ $\frac{d}{dx}(\csc x) = -\csc x \cot x$ $\frac{d}{dx}(\cot x) = -\csc^2 x$

INTEGRALS

Trig Functions

$\int \sin u\,du = -\cos u + c$ $\int \cos u\,du = \sin u + c$ $\int \tan u\,du = \ln|\sec u| + c$

$\int \sec u\,du = \ln|\sec u + \tan u| + c$ $\int \csc u\,du = \ln|\csc u - \cot u| + c$ $\int \cot u\,du = \ln|\sin u| + c$

$\int \sec^2 u\,du = \tan u + c$ $\int \csc^2 u\,du = -\cot u + c$

Trigonometric Substitutions

$a^2 - x^2 \Rightarrow x = \sin(\theta)$ or $x = \cos(\theta)$
$x^2 + a^2 \Rightarrow x = \tan(\theta)$
$x^2 - a^2 \Rightarrow x = \sec(\theta)$

TRIG IDENTITIES

Reciprocal Identities

$\sin\theta = \frac{1}{\csc\theta}$ $\quad \cos\theta = \frac{1}{\sec\theta}$ $\quad \tan\theta = \frac{1}{\cot\theta}$

$\csc\theta = \frac{1}{\sin\theta}$ $\quad \sec\theta = \frac{1}{\cos\theta}$ $\quad \cot\theta = \frac{1}{\tan\theta}$

Quotient Identities

$\tan\theta = \frac{\sin\theta}{\cos\theta}$ $\quad \cot\theta = \frac{\cos\theta}{\sin\theta}$

Pythagorean Identities

$1 = \sin^2\theta + \cos^2\theta$ $\quad \sec^2\theta = \tan^2\theta + 1$ $\quad \csc^2\theta = 1 + \cot^2\theta$

Double Angle Identities

$\sin 2\theta = 2\sin\theta\cos\theta$ $\quad \tan 2\theta = \frac{2\tan\theta}{1-\tan^2\theta}$ $\quad \cos 2\theta = 1 - 2\sin^2\theta$

Power Reducing Identities

$\sin^2\theta = \frac{1-\cos 2\theta}{2}$ $\quad \cos^2\theta = \frac{1+\cos 2\theta}{2}$ $\quad \tan^2\theta = \frac{1-\cos 2\theta}{1+\cos 2\theta}$

Partial Fractions

$$\int \frac{f(x)}{g(x)}\,dx$$

- the degree of $f(x)$ is less than the the degree of $g(x)$

- Need to factorize $g(x)$

Some examples:

$$\frac{px+q}{(x-a)(x-b)} = \frac{A}{(x-a)} + \frac{B}{(x-b)}, \quad a \neq b$$

$$\frac{px+q}{(x-a)^2} = \frac{A}{x-a} + \frac{B}{(x-a)^2}$$

$$\frac{px^2+qx+r}{(x-a)(x-b)(x-c)} = \frac{A}{x-a} + \frac{B}{x-b} + \frac{C}{x-c}$$

$$\frac{px^2+qx+r}{(x-a)(x-b)^2} = \frac{A}{x-a} + \frac{B}{x-b} + \frac{C}{(x-b)^2}$$

Partial Fractions

$$\int \frac{f(x)}{g(x)}\,dx$$

- the degree of $f(x)$ is less than the degree of $g(x)$
- Need to factorize $g(x)$

Some examples:

$$\frac{px+q}{(x-a)(x-b)} = \frac{A}{(x-a)} + \frac{B}{(x-b)}, \quad a \neq b$$

$$\frac{px+q}{(x-a)^2} = \frac{A}{x-a} + \frac{B}{(x-a)^2}$$

$$\frac{px^2+qx+r}{(x-a)(x-b)(x-c)} = \frac{A}{x-a} + \frac{B}{x-b} + \frac{C}{x-c}$$

$$\frac{px^2+qx+r}{(x-a)(x-b)^2} = \frac{A}{x-a} + \frac{B}{x-b} + \frac{C}{(x-b)^2}$$

$$\frac{px^2+qx+r}{(x-a)(x^2+bx+c)} = \frac{A}{x-a} + \frac{Bx+C}{x^2+bx+c}$$

This is an irreducible quadratic